

User guide for FltRup_haz (version 8)

-----output file names -----

line 1: header line

line 2: output file (a80) for mean hazard results (unit 60)

line 3: output file (a80) for hazard fractiles (unit 30)

line 4: output file (a80) for Disaggregation (unit 61)

-----logic tree for FDM -----

line 5: header line

line 6: nFDM1 (integer), number of FDMs

Repeat line 7 for each FDM

line 7: index_FDM_med, deltaMu1, deltaMu2, index_FDM_sigma, wt1

index_FDM_med: see list in Table 1.

deltaMu1 = constant ($m^{0.3}$) added to FDM median (e.g., scaled backbone term)

deltaMu2 = delta_NE_med ($m^{0.3}$) added to FDM median

index_FDM_sigma (central value): see list in Table 1.

wt1: branch weight for this FDM branch

if index_FDM_sigma = 0, then read the fixed value at the end of line 7

revised line 7: index_FDM_med, deltaMu1, deltaMu2, index_FDM_sigma, wt1, fixed_sig
($m^{0.3}$ units)

fixed_sig is a fixed value of sigma set by the user

-----logic tree for non-ergodic term, delta_NE-----

line 8: header line

line 9: nFDM2 (integer), number of epistemic branches for the delta_NE

line 10: deltaMu3, wt2

deltaMu3 = constant ($m^{0.3}$) epistemic unc in Delta_NE

wt2 = branch weight for delta_NE

note: repeat line 10 for each epistemic branch of the non-ergodic term

-----logic tree for sigma model adjustment -----

line 11: header line

line 12: nSig, number of epistemic unc in sigma

line 13: delta_sig, wt3

delta_sig = adjustment to the sigma model

wt3: branch weight for delta_sig

Note: repeat line 13 for each sigma adjustment

----- probability of surface rupture model -----

line 14: header line

line 15: nPSR , number of probability of surface rupture models

line 16: index_PSR, wt, c1, c2

index_PSR, see table 2 for list of PSR models

wt2 = branch weight for PSR model

c1 and c2 are the coeff for a logistic model for mag only

The c1 and c2 are only read if index_PSR=1

Note: repeat line 16 for each PSR model

----- surface rupture length model -----

line 17: header line

line 18: nRLM , number of rupture length models

line 19: index_RLM, wt

index_RLM, see table 3 for list of RL models

wt3 = branch weight for RL model

Note: repeat line 19 for each RL model

----- SSC inputs for magnitudes and rates -----

line 20: header line

line 21: input file (a80) for the out2 file from Haz45)

----- SSC inputs for SOF -----

line 22: header line

line 23: nSOF , number of SOF branches (epistemic)

line 24: iSOF, wt

iSOF= index for style-of-faulting

-1 = NML, 0 =SS, 1=RV

wt = branch weight for SOF

Note: repeat line 21 for each SOF model

----- Fault Geometry -----

line 25: header line

line 26: nWidth , number of seismogenic widths (measured vertical)

line 27: L_Flt, width, dip, wt

L_Flt = fault length in km

Width = seismogenic width (km), not the down-dip fault width

dip = fault dip (degrees)

wt = branch wt for the seismogenic width

Note: repeat line 21 for each width

----- site location -----

line 28: header line

line 29: XL_F, Rx

XL_F = X_F/L_F for the site relative to the fault (not to a rupture)

Line 30: L_site (m), W_site (m)

L_site = site dimension (m) measured along strike

W_site = site dimension (m) measured perpendicular to strike

----- z values for hazard calc -----

line 31: header line

line 32: nZ, number of disp values for the hazard calc

line 33: disp values (m) for hazard calc

----- disaggregation bins for hazard calc -----

line 34: header line

line 35: MagBinStep, nMagBin, Mag0

line 36: epsBinStep, nEpsBin, Eps0

line 37: XLBinStep, nXLBin, XL0

The first bin is ($Mag0 < Mag < Mag0 + MagBinStep$)

Table 1. FDM models

index	FDM	Disp Metric	Parameter
1	LA23	Sum of Principal Disp, Segmented	Median and sigma ($m^{0.3}$)
2	LA23	Sum of Principal Disp, Unsegmented	Median and sigma ($m^{0.3}$)
3	LA23	Aggregate Disp, Segmented	Median and sigma ($m^{0.3}$)
4	LA23	Aggregate Disp, Unsegmented	Median and sigma ($m^{0.3}$)
5	LA25	Principal Disp	Non-ergodic sigma ($m^{0.3}$) based on sigmaMag=0.1
6	LA25	Principal Disp	Non-ergodic sigma ($m^{0.3}$) based on sigmaMag=0.15

Table 2. Probability of surface rupture models

index	PSR model
0	PSR = 1
1	user specified logistic model coeff
2	
3	Wells and Coppersmith (1984)
4	Huang and Abrahamson (2026) median alpha
5	Huang and Abrahamson (2026) 5th alpha
6	Huang and Abrahamson (2026) 95th alpha

Table 2. Surface rupture length models

index	Surface Rupture Length model
1	Lavrentidias et al (2023)
2	
3	